

DEPARTMENT OF MATHEMATICS
CIE MODEL QUESTION PAPER

Sub Code:	CS/IS/CI/CY/Al/AD41	Sub:	Numerical Techniques and Probability Models	Test:	01
Semester:	IV	Term:	08-05-2023 to 12-08-2023	Marks:	30

Note: Answer any TWO full questions. Each main question carries 15 marks

Q.No.		Questions	Blooms Level	CO's	Marks																				
1.	(a)	Is divided difference is applicable for equidistant data? Justify	L1	CO1	2																				
	(b)	Fit an interpolating polynomial of degree three which takes the following values: <table><tr><td>x</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>6</td><td>24</td><td>60</td><td>120</td></tr></table>	x	3	4	5	6	y	6	24	60	120	L2	CO1	3										
x	3	4	5	6																					
y	6	24	60	120																					
	(c)	The following table gives agricultural production index (x) and the index of whole sale prices (y) in 5 years <table><tr><td>x</td><td>104</td><td>110</td><td>112</td><td>114</td><td>120</td></tr><tr><td>y</td><td>106</td><td>116</td><td>140</td><td>175</td><td>173</td></tr></table> (i) Find 'r', (ii) Find regression equation of wholesale price index on the agricultural production index. (iii) Obtain an estimate of wholesale price index when the agricultural production index is 125.	x	104	110	112	114	120	y	106	116	140	175	173	L3	CO2	5								
x	104	110	112	114	120																				
y	106	116	140	175	173																				
	(d)	A river is 80 feet wide. The depth 'd' at a distance 'x' from one bank is given by <table><tr><td>x(ft)</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td></tr><tr><td>d(ft)</td><td>0</td><td>4</td><td>7</td><td>9</td><td>12</td><td>15</td><td>14</td><td>8</td><td>3</td></tr></table> Find approximately the area of cross section of river.	x(ft)	0	10	20	30	40	50	60	70	80	d(ft)	0	4	7	9	12	15	14	8	3	L5	CO1	5
x(ft)	0	10	20	30	40	50	60	70	80																
d(ft)	0	4	7	9	12	15	14	8	3																
2.	(a)	When can Simpson's 1/3 rd rule be applicable to the integral. Explain?	L1	CO1	2																				
	(b)	Write normal equation for $y = a - \frac{b}{x^2}$ by the method of least square.	L2	CO2	3																				
	(c)	Let $y(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-x^2} dx$ be the integral , Given <table><tr><td>x</td><td>0.46</td><td>0.47</td><td>0.48</td><td>0.49</td></tr><tr><td>y</td><td>0.4846</td><td>0.4937</td><td>0.5027</td><td>0.5116</td></tr></table> Determine for what value 'x' $y(x) = \frac{1}{2}$.	x	0.46	0.47	0.48	0.49	y	0.4846	0.4937	0.5027	0.5116	L3	CO1	5										
x	0.46	0.47	0.48	0.49																					
y	0.4846	0.4937	0.5027	0.5116																					
	(d)	The following table contains the path $y = f(t)$ traversed in time 't' by a moving particle. Find velocity and acceleration at the point of time 't =0.04' <table><tr><td>t</td><td>0</td><td>0.01</td><td>0.02</td><td>0.03</td><td>0.04</td></tr><tr><td>y</td><td>0</td><td>1.519</td><td>6.031</td><td>23.396</td><td>35.721</td></tr></table>	t	0	0.01	0.02	0.03	0.04	y	0	1.519	6.031	23.396	35.721	L4	CO1	5								
t	0	0.01	0.02	0.03	0.04																				
y	0	1.519	6.031	23.396	35.721																				
3.	(a)	Are regression lines of 'y' on 'x' and 'x' on 'y' for a given data same? Give reason.	L1	CO2	2																				
	(b)	Using Trapezoidal rule evaluate $\int_1^{10} \frac{dx}{x}$ by taking 10 ordinates.	L2	CO1	3																				
	(c)	Estimate the chlorine residual in a swimming pool 5 hours after it has been treated with chemicals, by fitting an exponential curve of the form $y = ab^x$ to the following data. <table><tr><td>Number of hours (x)</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td></tr><tr><td>Chlorine Residual parts (y)</td><td>1.8</td><td>1.5</td><td>1.4</td><td>1.1</td><td>1.1</td><td>0.9</td></tr></table>	Number of hours (x)	2	4	6	8	10	12	Chlorine Residual parts (y)	1.8	1.5	1.4	1.1	1.1	0.9	L3	CO2	5						
Number of hours (x)	2	4	6	8	10	12																			
Chlorine Residual parts (y)	1.8	1.5	1.4	1.1	1.1	0.9																			
	(d)	Given the following score distribution in mathematics. Find the number of students who scored marks between 45 and 55 <table><tr><td>Marks</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td></tr><tr><td>No.of students</td><td>52</td><td>36</td><td>21</td><td>20</td><td>15</td></tr></table>	Marks	30-40	40-50	50-60	60-70	70-80	No.of students	52	36	21	20	15	L4	CO1	5								
Marks	30-40	40-50	50-60	60-70	70-80																				
No.of students	52	36	21	20	15																				

DEPARTMENT OF MATHEMATICS, QUIZ QUESTION PAPER

Sub Code: CS/IS/CY/CI/AI/AD41

Total marks: 10M

Note: Answer all questions. Each question carries 2 marks

1MS21AD029

Q.No.	Questions
1.	If $f(x) = x^2$, then second order divided difference for the point x_0, x_1, x_2 is ?
2.	Following are the values of the function $y(x)$, $y(-1) = 5, y(0) = 0, y(1) = 8$. Find y' at $x = 0$.
3.	A person throws a fair coin. He gets Rs. 8/- if head appears. Otherwise, he loses Rs. 5/-. Find his expectation. Also, find the variance.
4.	Out of the following two regression lines $2x + 3y = 7, 5x + 4y = 9$, which one will be regression line of x on y ? Explain?
5.	The average number of telephone calls booked at an exchange between 10-00 A. M. and 10-10 A. M. is 4. Find the probability that on a randomly selected day 2 or more calls are booked between 10-00 A. M. and 10-10 A. M. On how many days of a leap year, would you expect booking of 2 or more calls during that time gap?



DEPARTMENT OF MATHEMATICS

Sub Code:	AD41/AI41	Sub:	Statistics and Probability for Data science	Test:	I
Time:	2.45pm to 3.45 p.m	Term:	15.04.24 to 27.07.24	Marks:	30
Date:	12.07.24	Semester:	IV	Sections:	AD&AI

Note: Answer any TWO full questions. Each main question carries 15 marks.

Answer any TWO full questions. Each main question carries 15 marks.

Q.No.	Questions	Blooms level	CO's	Marks																						
1.	(a) Fit a least square geometric curve $y = ax^b$ to the following data. <table border="1"> <tr> <td>x</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> </tr> <tr> <td>y</td> <td>0.5</td> <td>2</td> <td>4.5</td> <td>8</td> <td>12.5</td> </tr> </table>	x	1	2	3	4	5	y	0.5	2	4.5	8	12.5	L1	CO1	5										
x	1	2	3	4	5																					
y	0.5	2	4.5	8	12.5																					
	(b) Ten students got the following percentage of marks in in both the laboratory and lecture portions of Biology course. <table border="1"> <tr> <td>laboratory</td> <td>8</td> <td>3</td> <td>9</td> <td>2</td> <td>7</td> <td>10</td> <td>4</td> <td>6</td> <td>1</td> <td>5</td> </tr> <tr> <td>Lecture</td> <td>9</td> <td>5</td> <td>10</td> <td>1</td> <td>8</td> <td>7</td> <td>3</td> <td>4</td> <td>2</td> <td>6</td> </tr> </table> Calculate the rank correlation coefficient.	laboratory	8	3	9	2	7	10	4	6	1	5	Lecture	9	5	10	1	8	7	3	4	2	6	L3	CO1	5
laboratory	8	3	9	2	7	10	4	6	1	5																
Lecture	9	5	10	1	8	7	3	4	2	6																
	(c) If $r_{12} = \frac{1}{2}$, $r_{23} = \frac{3}{4}$, $r_{31} = \frac{2}{3}$ then find the values of $R_{1.23}$ and $r_{1.32}$.	L4	CO2	5																						
2.	(a) If the coefficient of correlation between two variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}\left(\frac{3}{5}\right)$. Show that $\sigma_y = 2\sigma_x$ or $\sigma_x = 2\sigma_y$.	L1	CO2	5																						
	(b) Fit a parabola of second degree $y = a + bx + cx^2$ for the following data <table border="1"> <tr> <td>x</td> <td>1.2</td> <td>1.8</td> <td>3.1</td> <td>4.9</td> <td>5.7</td> <td>7.1</td> <td>8.6</td> <td>9.8</td> </tr> <tr> <td>y</td> <td>4.5</td> <td>5.9</td> <td>7.0</td> <td>7.8</td> <td>7.2</td> <td>6.8</td> <td>4.5</td> <td>2.7</td> </tr> </table>	x	1.2	1.8	3.1	4.9	5.7	7.1	8.6	9.8	y	4.5	5.9	7.0	7.8	7.2	6.8	4.5	2.7	L2	CO1	5				
x	1.2	1.8	3.1	4.9	5.7	7.1	8.6	9.8																		
y	4.5	5.9	7.0	7.8	7.2	6.8	4.5	2.7																		
	(c) In a partially destroyed laboratory record of an analysis of correlation data, the following results only are legible: Variance of $x = 9$. Regression equations are : $8x - 10y + 66 = 0$, $40x - 18y = 214$. Find (i) the mean of x and y. (ii) correlation between x and y (iii) the standard deviation of y.	L5	CO2	5																						

3.	(a)	Calculate the correlation coefficient for the following age of employees(x) and the number of days(y) they were reported sick in a month. <table><tr><td>x</td><td>30</td><td>32</td><td>35</td><td>40</td><td>48</td><td>50</td><td>52</td><td>55</td><td>57</td><td>61</td></tr><tr><td>y</td><td>1</td><td>0</td><td>2</td><td>5</td><td>2</td><td>4</td><td>6</td><td>5</td><td>7</td><td>8</td></tr></table>	x	30	32	35	40	48	50	52	55	57	61	y	1	0	2	5	2	4	6	5	7	8	L1	CO1	5
	x	30	32	35	40	48	50	52	55	57	61																
	y	1	0	2	5	2	4	6	5	7	8																
(b)	For the following data find the regression lines of y on x and x on y . <table><tr><td>x</td><td>6</td><td>2</td><td>10</td><td>4</td><td>8</td></tr><tr><td>y</td><td>9</td><td>11</td><td>5</td><td>8</td><td>7</td></tr></table>	x	6	2	10	4	8	y	9	11	5	8	7	L4	CO2	5											
x	6	2	10	4	8																						
y	9	11	5	8	7																						
(c)	Find Y when $X_1=15$, $X_2=8$ from the least square regression equation of Y on X_1 and X_2 for the following data. <table><tr><td>Y</td><td>90</td><td>72</td><td>54</td><td>42</td><td>30</td><td>12</td></tr><tr><td>X_1</td><td>3</td><td>5</td><td>6</td><td>8</td><td>12</td><td>14</td></tr><tr><td>X_2</td><td>16</td><td>10</td><td>7</td><td>4</td><td>3</td><td>2</td></tr></table>	Y	90	72	54	42	30	12	X_1	3	5	6	8	12	14	X_2	16	10	7	4	3	2	L3	CO2	5		
Y	90	72	54	42	30	12																					
X_1	3	5	6	8	12	14																					
X_2	16	10	7	4	3	2																					



DEPARTMENT OF MATHEMATICS

Sub Code:	CS/IS/AI/AD/CY/CI41	Sub:	Statistics, Probability and Linear Programming		Test:	02
Time :	3: 00 pm To 04.00 pm	Term:	03:03:2025 To 21-06-2025		Marks:	30
Date:	16.06.2025	Semester:	IV	Section:	All	

Note: Answer any **TWO** full questions. Each question carries 15 marks

Q. No.	Questions		Blooms Level	CO's	Marks									
1.	(a)	Define Absorbing state	L1	CO3	2									
	(b)	Write a short note on Type I and Type II errors	L2	CO4	3									
	(c)	Solve using graphical method, $\min z = 200x + 500y$, subject to the constraints $x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$.	L5	CO5	5									
	(d)	A supermarket has a single cashier. During peak hours, customers arrive at a rate of 20 per hour. The average number of customers that can be processed by the cashier is 24 per hour. Calculate i)The average number of customers in the queuing system ii)The average time a customer spends in the system iii)The average number of customers in the queue	L4	CO3	5									
2.	(a)	Write the expression to determine the expected waiting time of a customer in the M/M/1 system	L1	CO3	2									
	(b)	A manufacturing company makes two types of television sets; one is black and white and the other is colour. The company has resources to make at most 300 sets a week. It takes Rs 1800 to make a black and white set and Rs 2700 to make a colored set. The company can spend no more than Rs 648000 a week to make television sets. If it makes a profit of Rs 510 per black and white set and Rs 675 per colored set, how many sets of each type should be produced so that the company has a maximum profit? Formulate this problem as a LPP given that the objective is to maximize the profit.	L2	CO5	3									
	(c)	There are 2 white marbles in bag A and 3 red marbles in bag B. At each step of the process a marble is selected at random from each bag and the two marbles selected are interchanged then find (i) the transition probability matrix (ii) what is the probability that there are 2 red marbles in A after 3 steps	L3	CO3	5									
	(d)	A fertilizer mixing machine is set to give 12 kg of nitrate for quintal bag of fertilizer. Ten 100 kg bags are examined, the percentage of nitrate per bag are as follows: 11, 14, 13, 12, 13, 12, 13, 14, 11, 12. Are there any reasons to believe that the machine is defective?	L4	CO4	5									
3.	(a)	Is the stochastic matrix $\begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$ regular? Justify.	L1	CO3	2									
	(b)	Find Minimum $z = 5x + 4y$ subject to the constraints $x + 2y \leq 10$, $x + y \geq 8$, $2x + y \geq 12$, $x \geq 0$, $y \geq 0$ by graphical method.	L2	CO5	3									
	(c)	Patients arrive at a doctor's clinic according to Poisson distribution at a rate of 30 patients per hour. The waiting room does not accommodate more than 9 patients. Examination time per patient is exponential with mean rate of 20 per hour. Find the: i) Probability that an arriving patient will not wait ii) Average number of patients in the clinic iii) Expected time a patient spends in the clinic	L5	CO3	5									
	(d)	The following table shows the result of an experiment to investigate the effect of vaccination induced on the animals against a particular disease. Use χ^2 - test to test the hypothesis that there is no difference between and vaccinated and unvaccinated groups i.e. vaccination and this disease are independent	L4	CO4	5									
		<table><tr><td></td><td>Got disease</td><td>Did not get disease</td></tr><tr><td>vaccinated</td><td>9</td><td>42</td></tr><tr><td>Not vaccinated</td><td>17</td><td>28</td></tr></table>		Got disease	Did not get disease	vaccinated	9	42	Not vaccinated	17	28			
	Got disease	Did not get disease												
vaccinated	9	42												
Not vaccinated	17	28												



DEPARTMENT OF MATHEMATICS

Sub Code:	CS/IS/AI/AD/CY/CI41	Sub:	Statistics, Probability and Linear Programming	Test:	01
Time :	3: 00 pm To 04.00 pm	Term:	03:03:2025 To 21-06-2025	Marks:	30
Date:	24.04.2025	Semester:	IV	Section:	All

Note: Answer any **TWO** full questions. Each question carries 15 marks

Q. No.	Questions	Blooms Level	CO's	Marks																					
1. (a)	Write normal equations to fit a curve of the form $y = \frac{a}{x^3} - b$.	L1	CO1	2																					
(b)	Suppose the time it takes a student to finish a quiz is uniformly distributed between six and 15 minutes. Find the probability that a randomly selected student needs at least eight minutes to complete the quiz. Then find the probability that a different student needs at least eight minutes to finish the quiz given that she has already taken more than seven minutes.	L2	CO2	3																					
(c)	Two cards are selected at a random from a box which contains seven cards numbered 1,1,1,2,2,2 and 3. Find the joint distribution of X and Y, where X denotes the sum and Y the maximum of the two numbers drawn. Also determine $COV(X, Y)$.	L5	CO2	5																					
(d)	Find Y when $X_1 = 10$, $X_2 = 6$ from the least square regression equation of Y on X_1 and X_2 for the following data: <table><tr><td>Y</td><td>90</td><td>72</td><td>54</td><td>42</td><td>30</td><td>12</td></tr><tr><td>X_1</td><td>3</td><td>5</td><td>6</td><td>8</td><td>12</td><td>14</td></tr><tr><td>X_2</td><td>16</td><td>10</td><td>7</td><td>4</td><td>3</td><td>2</td></tr></table>	Y	90	72	54	42	30	12	X_1	3	5	6	8	12	14	X_2	16	10	7	4	3	2	L4	CO1	5
Y	90	72	54	42	30	12																			
X_1	3	5	6	8	12	14																			
X_2	16	10	7	4	3	2																			
2. (a)	Define discrete random variable with an example	L1	CO1	2																					
(b)	Suppose that the longevity of a light bulb is exponential with a mean lifetime of eight years. i) Find the probability that a bulb chosen at random last over 19 years. ii) If a bulb has already lasted 12 years, find the probability that it will last a total of over 19 years.	L2	CO2	3																					
(c)	Out of 888 families having 4 children each, in how many families would you expect at least 2 boys. Given that the probability of a boy is twice that of girl.	L3	CO1	5																					
(d)	Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 527 and a standard deviation of 112. What is the probability of an individual scoring above 500 on the GMAT? How high must an individual score on the GMAT in order to score in the highest 5%? If 5 students are selected at random what is the probability that at least one is on the top 5%.	L4	CO2	5																					
3. (a)	Write the pdf of Gamma probability distribution	L1	CO2	2																					
(b)	On average, the surface area A of human beings is related to weight W and height H. Measurements on a number of individuals of height 180 cm and different weights (kg) give values of A (m^2) in the following table. <table><tr><td>W (kg)</td><td>70</td><td>75</td><td>77</td><td>80</td><td>82</td><td>84</td><td>87</td><td>90</td></tr><tr><td>A (m^2)</td><td>2.10</td><td>2.12</td><td>2.15</td><td>2.20</td><td>2.22</td><td>2.23</td><td>2.26</td><td>2.30</td></tr></table> Fit a curve of the form $A = aW^b$. Also, predict the surface area for a person who weighs 95 kg.	W (kg)	70	75	77	80	82	84	87	90	A (m^2)	2.10	2.12	2.15	2.20	2.22	2.23	2.26	2.30	L2	CO1	3			
W (kg)	70	75	77	80	82	84	87	90																	
A (m^2)	2.10	2.12	2.15	2.20	2.22	2.23	2.26	2.30																	
(c)	The pdf of a random variable X is given by $f(x) = \begin{cases} ax & 0 \leq x \leq 1 \\ a & 1 \leq x \leq 2 \\ 3a - ax & 2 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$. Find the cdf and hence find $P[X \geq 1.5]$.	L5	CO2	5																					
(d)	An airline knows that 8% of the people making reservations will not turn up. Consequently, their policy is to sell 111 tickets for a flight that can hold only 101 passengers. What is the probability that every passenger who turns up will have a seat if it's given that at least 95 passengers will turn up?	L4	CO1	5																					

QUIZ TEST FOR CS(AIML) SECTION-B

SUBJECT: STATISTICS PROBABILITY AND LINEAR PROGRAMMING

Answer all questions

Time: 30 mins

Marks $5 \times 2 = 10$

- 1 A continuous random variable has a pdf $f(x) = 3x$, $1 \leq x \leq 2$. Find a such that $P(X \leq a) = P(X > a)$
- 2 In a sample of 1000 cases, the mean of a certain test is 14 and standard deviation is 2.5. Assuming the distribution to be normal, find how many students score between 12 and 15?
- 3 Find the fixed probability vector of the matrix $\begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix}$
- 4 Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate of 20 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer.
- 5 Find the joint distribution of X and Y , Which are independent random variables with the following respective distributions, also find $E(X)$.

x_i	1	2
$f(x_i)$	0.7	0.3

y_i	-2	5	8
$g(y_i)$	0.3	0.5	0.2



SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JULY / AUGUST 2025

Program	: B.E. - Common to CSE / ISE / AI & ML / AI & DS / CSE (AI & ML) / CSE (CY)	Semester	: IV
Course Name	: Statistics, Probability and Linear Programming	Max. Marks	: 100
Course Code	: 23 CS / IS / AI / AD / CI / CY 41 22 CS / IS / CI / CY41	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Use of Statistical Table is permitted.

UNIT - I

- Define Continuous random variable and give an example. CO1 (02)
 - Out of 520 families with 5 children each, how many would you expect to have (i) 3 boys (ii) 3 girls. Assume the probability of a girl child is 0.45. CO1 (04)
 - A random variable X has the following probability function: CO1 (07)

X	1	2	3	4	5	6	7
$P(X)$	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Find (i) k (ii) $P(X \geq 6)$ (iii) $E(X)$.

- Fit a least square regression plane $Y = a + bX_1 + cX_2$ for the following data: CO1 (07)

Y	9	7	5	4
X_1	30	52	64	82
X_2	16	10	7	4

- Write the equations of two regression lines. CO1 (02)
 - A continuous random variable X has pdf given by $f(x) = \begin{cases} 2e^{-2x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$ CO1 (04)

Evaluate (i) $E(X)$ (ii) $E(X^2)$.

- Dynamic viscosity of water $\mu (10^{-3} \text{ N.s/m}^2)$ is related to temperature $T (^{\circ}\text{C})$ in the following manner. Estimate viscosity of water at $T = 35^{\circ}\text{C}$ by framing a parabolic relation $\mu = aT^2 + bT + c$. CO1 (07)

T	0	5	10	20	30	40
μ	1.787	1.519	1.307	1.002	0.7975	0.6529

- A communication channel receives independent pulses at the rate of 12 pulses per micro second. The probability of transmission error is 0.001 for each micro second. Compute the probability of (i) no error during a micro second (ii) one error per micro second (iii) at least three error per micro second (iv) At most two errors. CO1 (07)

UNIT - II

- Write pdf of Gamma distribution. CO2 (02)
 - On a certain city transport route, buses ply every 30 minutes between 6 a.m. and 10 p.m. If a person reaches a bus stop on this route at a random time during this period, what is the probability that he will have to wait for at least twenty minutes? CO2 (04)

- c) The joint distribution of two variables X and Y are given below

CO2 (07)

$X \backslash Y$	-4	2	7
1	1/8	1/4	1/8
5	1/4	1/8	1/8

Find (i) Marginal distributions of X and Y (ii) $COV(X, Y)$

- d) A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live (i) more than 32 months (ii) less than 28 months (iii) between 37 and 49 months. CO2 (07)
4. a) Write mean and variance of uniform distribution. CO2 (02)
- b) Steel rods are manufactured to be 3 cm in diameter but they are acceptable if they are inside the limits 2.99 cm and 3.01 cm. It is observed that 5% are rejected as oversized and 5% are rejected as undersized. Assuming that the diameters are normally distributed, find the standard deviation of the distribution. CO2 (04)
- c) The mileage which car owner get with a certain kind of radial tyre is a random variable having an exponential distribution with mean 30,000km. Find the probabilities that one of these tyres will last (i) at least 15,000km (ii) at most 25,000km. CO2 (07)
- d) A privately owned business operates both drive in facility and a walk in facility. On a randomly selected day let X and Y respectively be the proportions of the time that drive in and walk in facilities are in use and suppose that the joint density function of these random variables is CO2 (07)

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Show that (i) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$ (ii) Find $P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right)$.

UNIT - III

5. a) Write the expression for i) expected number of customers in the queue ii) expected number of customers in the systems for (M/M/s):(∞/FIFO) queuing system. CO3 (02)
- b) Suppose that customers arrive at a Poisson rate of one per every 12 minutes and that the service time is exponential at a rate of one service per 8 minutes. What is the average number of customers in the system. CO3 (04)
- c) There are 2 white marbles in bag A and 3 red marbles in bag B. At each step of the process a marble is selected at random from each bag and the two marbles selected are interchanged then find (i) transition probability matrix (ii) what is the probability that there are 2 red marbles in A after 3 steps (iii) In the long run what is the probability that there are 2 red marbles in A? CO3 (07)
- d) Trains arrive at the yard every 15 minutes and the service time is 33 minutes. If the line capacity of the yard is limited to 4 trains, then find CO3 (07)
- (i) The probability that the yard is empty.
- (ii) The average number of trains in the system.

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6. a) Write the expression to determine the expected number of customers in the queue for (M/M/1): (k/FIFO). CO3 (02)
- b) Check whether the following Stochastic matrix is Regular or not? CO3 (04)

$$\begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$$

- c) Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate of 20 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer in the queue and in the system. CO3 (07)
- d) A petrol pump has 4 pumps. The service time follows an exponential distribution with a mean of 6 minutes and cars arrive for service in a Poisson process at the rate of 30 cars per hour. Find the average waiting time in the queue, average time spent in the system and the average number of cars in the system. CO3 (07)

UNIT- IV

7. a) Define Null Hypothesis and Alternative Hypothesis. CO4 (02)
- b) The nine items of a sample have the following values: 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these differ significantly from the assumed population mean of 47.5? Test at 5% LOS. CO4 (04)
- c) A sample of size 13 gave an estimated population variance of 3.0, while another sample of size 15 gave an estimate of 2.5. Could both samples be from populations with the same variance? Test at 5% LOS. CO4 (07)
- d) The following table shows the result of an experiment to investigate the effect of vaccination induced on the animals against a particular disease. Use χ^2 - test to test the hypothesis that there is no difference between and vaccinated and unvaccinated groups i.e. vaccination and this disease are independent. Test at 5% LOS. CO4 (07)

	Got disease	Did not get disease
vaccinated	9	42
Not vaccinated	17	28

8. a) Define Type-I and Type - II errors. CO4 (02)
- b) A sample of 900 members is found to have a mean of 3.4 cm. Can it be reasonably regarded as a truly random sample from a large population with mean 3.25 cm and S.D 1.61 cm. Test at 1% LOS. CO4 (04)
- c) Two independent samples of sizes 8 and 7 contained the following values: CO4 (07)

Sample 1	19	17	15	21	16	18	16	14
Sample 2	15	14	15	19	15	18	16	

Is the difference between the sample means significant? Test at 5%LOS

- d) The theory predicts that the proportion of beans available in four groups A, B, C, D should be 9:3:3:1. In an experiment with 1600 beans the numbers in the four groups were 882, 313, 287 and 118. Use χ^2 - test to verify whether the experimental result supports the theory. Test at 5% LOS. CO4 (07)

UNIT - V

9. a) Use Big-M method to maximize $z = 3x_1 + 2x_2$, subject to the constraints, $2x_1 + x_2 \leq 2$, $3x_1 + 4x_2 \geq 12$, $x_1 \geq 0$, $x_2 \geq 0$. CO5 (10)
- b) Solve the following LPP by Simplex method : CO5 (10)
- $Max\ Z = 6x_1 + 11x_2$
 $s/t\ 2x_1 + x_2 \leq 104$
 $x_1 + 2x_2 \leq 76$
 $x_1, x_2 \geq 0$
10. a) Convert the following problem in dual form: CO5 (04)
- Maximize: $z = x + y + z$ subjected to $x - 3y + 4z = 5$, $x - 2y \leq 3$,
 $2y - z \geq 4$, where $x, y \geq 0, z$ unrestricted.
- b) Solve $Min\ z = 5x + 4y$ subject to the constraints $x + 2y \leq 10$, $x + y \geq 8$, CO5 (06)
- $2x + y \geq 12$, $x \geq 0$, $y \geq 0$ by graphical method.
- c) Solve the following LPP by Two-phase method CO5 (10)
- $Min\ Z = 2x_1 + 3x_2$
 $s/t\ x_1 + x_2 \leq 5$
 $x_1 + 2x_2 \geq 6$
 $x_1, x_2 \geq 0$
